



CROWN

Note:

The technical data mentioned may be updated or revised due to product development.

The data in this brochure is subject to change without notice.

The latest datasheet and catalogue can be acquired via info@crownmicroglobal.com

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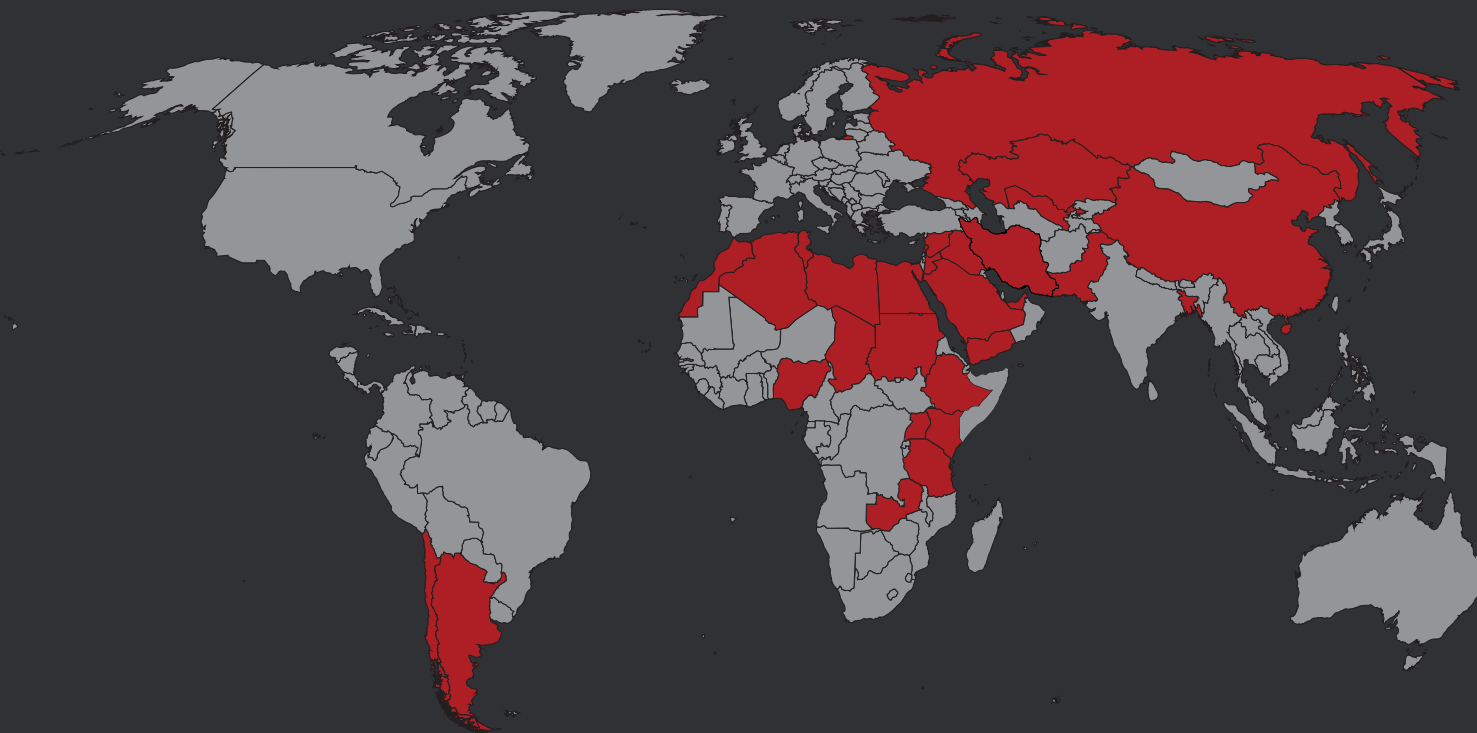


PRODUCT CATALOG

Power You Can Trust

CROWN

Delivering Reliable Energy Storage & Solar Solutions Across the Globe



Operating in more than 30 countries and expanding...

MISSION

Delivering superior power products with quality, innovation, and customer satisfaction at the core.

VISION

Quality, Innovation, Customer focus, Sustainability.

VALUE

Driving global progress with reliable, efficient, and sustainable energy solutions.



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 - HIGH VOLTAGE BATTERY RACK

Single Phase Off-Grid Inverter



1.5KW
3KW



-  Easy installation, one battery 12VDC (1.5KW)
-  Smart Battery Equalization
- BMS** Built-in BMS for 3kW
-  Built-in Wi-fi (3KW)
-  Anti Dust Kit
-  VDC Range 30~400
-  Max Efficiency 93%
- PF1 Output power factor 1
-  2.5" LCD with touch pad



MODEL NAME	XAVIER-1.5KW-PF1	XAVIER-3KW-PF1
RATED POWER	1500VA/1500W	3000VA/3000W
INPUT		
Voltage	230VAC	
Selectable voltage range	170 ~ 280VAC (for personal computers) 90 ~ 280VAC (for home appliances)	
Frequency range	50 Hz / 60 Hz (auto sensing)	
OUTPUT		
AC voltage regulation (batt. mode)	230VAC ± 5%	
Output power	1500W with PV & battery 1200W with battery only	3000W
Surge power	2400VA	6000VA
Efficiency (peak)	93%	
Transfer time	10ms (for personal computers)	
Waveform	Pure sine wave	
BATTERY		
Battery voltage	12VDC	24VDC
Floating charge voltage	13.5VDC	27VDC
Overcharge protection	16VDC	32VDC
Battery type	Lithium / Lead-acid	Lithium / Lead-acid
SOLAR CHARGER & AC CHARGER		
Solar charger type	MPPT	MPPT
Max. PV array open circuit voltage	400VDC	450VDC
Max. PV array power	2500W	3000W
MPPT range @ operating voltage	30 ~ 400VDC (30V ~ 60V with battery)	60 ~ 400VDC
Max. solar charge current	80A	100A
Max. AC charge current	80A	80A
Max. charge current	80A	100A
GENERAL SPECIFICATION		
Dimension, D*W*H (mm)	90*288*357	100*288*390
Net weight (Kgs)	6.6	7.5
Communication interface	RS232	
OPERATING ENVIRONMENT		
Humidity	5% to 95% relative humidity (non-condensing)	
Operating temperature	-10°C to 50°C	
Storage temperature	-15°C to 60°C	

Single Phase Hybrid Inverter



4KW
6KW



4KW
6KW



 Feed-in to the Grid	 Works Without Battery	BMS Battery Management System
 Built-in Wi-fi	 Dual Output	 VDC Range 60-450
 Max Efficiency 93%	 Built-in 120A Mppt solar charger	 5.2" LCD with touch pad

MODEL NAME	XAVIER-III-4KW	XAVIER-III-6KW
RATED POWER	4000VA/4000W	6000VA/6000W
INPUT		
Voltage	230VAC	
Selectable voltage range	170 ~ 280VAC (for personal computers) 90 ~ 280VAC (for home appliances)	
Frequency range	50 Hz / 60 Hz (auto sensing)	
OUTPUT		
AC voltage regulation (batt. mode)	230VAC ± 5%	
Surge power	7200VA	12000VA
Efficiency (peak)	93%	
Transfer time	10ms (for personal computers), 20ms (for home appliances)	
Waveform	Pure sine wave	
Dual outputs	Yes	
BATTERY		
Battery voltage	24VDC	48VDC
Floating charge voltage	27VDC	54VDC
Overcharge protection	33VDC	63VDC
SOLAR CHARGER & AC CHARGER		
Solar charger type	MPPT	
Max. PV array open circuit voltage	500VDC	
Max. PV array power	6000W	9000W
Max. PV input current	40A	
MPPT range @ operating voltage	60 ~ 450VDC	
Max. solar charge current	120A	
Max. AC charge current	100A	
Max. charge current	120A	
GENERAL SPECIFICATION		
Dimension, D*W*H (mm)	107*322.6*420.3	
Net weight (Kgs)	9.5	10.5
Communication interface	Wi-fi / RS-485 / CAN	
OPERATING ENVIRONMENT		
Humidity	5% to 95% relative humidity (non-condensing)	
Operating temperature	-10°C to 50°C	
Storage temperature	-15°C to 60°C	



PV - 6000 | PV - 9000



Single Phase Hybrid Inverter

ARCEUS 3.6KW
6.5KW

ARCEUS 3.6KW
6.5KW



 Feed-in to the Grid	 Works Without Battery	BMS Battery Management System
 Built-in Wi-Fi (6.5KW) Optional Wi-Fi (3.6KW)	 Dual Output	 VDC Range 90-550 (6.5KW)
 Parallel Operation up to 9 Units(6.5KW)	 Built-in 120A MPPT solar charger (6.5KW)	 5.2" LCD with touch pad



MODEL NAME	ARCEUS-3.6KW	ARCEUS-6.5KW
RATED POWER	3600VA/3600W	6500VA/6500W
AC INPUT		
Rated input voltage (VAC)	208 / 220 / 230 / 240; L + N + PE	
Voltage range (VAC)	90 ~ 280VAC $\pm$ 3V(App. mode),170 ~ 264VAC $\pm$ 3V(UPS mode)	
Frequency (Hz)	50 / 60 (auto adaptive)	
AC OUTPUT		
Rated capacity (KW)	3.6	6.5
Peak power (KVA)	3.6	6.5
Rated voltage (VAC)	208/220/230/240VAC $\pm$ 5%	
Power factor (PF)	1.0	
Frequency	50/60 Hz 10.1%	
Switch time (ms)	10 (App. mode) / 10 (UPS mode)	
Wave form	Pure sine wave	
Overload capacity (batt. mode)	10min @102% ~ 110% load,	1min @102% ~ 110% load,
	10S @ 110% ~ 130% load	10s @ 110% ~ 130% load,
	3S @ 130% ~ 150% load,	3s @ 130% ~ 150% load,
	200ms @ > 150% load	200ms@> 150% load
Max. efficiency (peak)	93.5% @ 24VDC	94.0%
Parallel quantity	N/A	4 units
CHARGER (PV / AC)		
Solar charger type	MPPT	
Max. PV input current / input power	18A / 5000W	27A /7800W
MPPT range@operating voltage (VDC)	40 ~ 450VDC	90 ~ 550VDC
Max. PV open circuit voltage (VDC)	500VDC	550VDC
Max. PV charge current (A)	100A	120A
Max. AC charge current (A)	100A	100A
Max. charge current (PV + AC) (A)	100A	100A
BATTERY		
Normal voltage (VDC)	24	48
Floating charge voltage (VDC)	27.6	54.0
Overcharge protection (VDC)	30	60
Battery type	Lithium / Lead-acid	Lithium / Lead-acid
INTERFACE		
HMI	5.2” LCD, display running mode/loads/input/output etc	
Interface	RS485 / CAN / USB / RS232 / dry contact	
Monitoring	Wi-fi (3.6 optional) / 6.5 (built-in)	
GENERAL SPECIFICATION		
Ingress protection	IP20	
Operating temperature	0°C to 40°C	
Relative humidity	20% to 95% (non-condensing)	
Storage temperature	-15°C to 60°C	
Net weight (Kg)	7.8	9.2
Dimensions, W*H*D (mm)	312*430*120	375*570*200
Max. operating altitude	4000m (derating above 1000m)	

Single Phase Hybrid Inverter



- Feed-in to the Grid
- Separate Generator Port Available
- BMS** Built-in BMS
- Built-in Wi-Fi
- Dual Output
- Touch Screen
- Dual MPPT PV Input
- Built-in 120A Mppt solar charger
- Parallel Operation up to 16 Units



PV - 9600



MODEL NAME	ELEGO-STAR-6.5KW
RATED POWER	6500VA/6500W
PV INPUT (DC)	
Max. PV array power (W)	9600
Rated PV input voltage (V)	320
Number of independent MPPT inputs	2
PV input voltage range (V)	100 ~ 500
MPPT voltage range (V)	120 ~ 400
Start-up voltage (V)	100
Max. PV input current per MPPT (A)	17 / 17
Max. PV short-circuit current input per MPPT (A)	25 / 25
BATTERY	
Compatible battery type	Lithium-ion / Lead-Acid
Rated battery voltage (V)	48
Battery voltage range (V)	38.4 ~ 60
Max. charge / discharge current (A)	140 / 140
Max. charge / discharge power (W)	6000/6500
Recommend capacity of battery per inverter	>200AH
Force wake up battery from PV function	Yes
Force wake up battery from Grid function	Yes
GRID	
Rated AC voltage (V)	230
Rated AC frequency (Hz)	50 / 60
Rated AC output current (A)	28.3
Rated AC output power (W)	6500
Max. AC input power (W)	9000
Power factor	0.99
THDI	<5%
Max. continuous AC passthrough current (A)	40
UPS	
Rated output power (W)	6500
Rated output voltage (V)	230
Rated output current (A)	28.3
Rated output frequency (Hz)	50 / 60
Surge power, duration	2Pn, <2s
Switching time (Ups)	<15ms @ single / <30ms @ parallel
Wave form	sine wave
THDV	3%
EFFICIENY	
MPPT efficiency	99.9%
Max. efficiency	93.0%
Max. charge efficiency	93.0%
Max. discharge efficiency	93.0%
PROTECTION	
Over current / voltage protection	YES
AC short - circuit current protection	YES
Grid monitoring	YES
DC surge protection type III	YES
AC surge protection type III	YES
Battery reverse polarity protection	YES
GENERAL	
Dimensions, W*H*D (mm)	330*505*135
Weight (Kgs)	14.5
Ingress protection rating	IP21
Operating environment temperature range (°C)	0 ~5 0
Storage temperature range (°C)	-15 ~ 60
Relative humidity	5% ~ 95%
Display communication interface	Touch screen, RS485 / Wifi / CAN
Cooling method	Smart cooling
Topology on AC side	Transformer-less
Altitude (M)	<2000m
Noise emission (typical)	<50dB

Single Phase Hybrid Inverter

NOVA 8.6KW
9.8KW

-  Feed-in to the Grid
-  Built-in Wi-Fi
-  Dual MPPT Input
-  Separate Generator Port Available
-  Dual Output
-  Built-in 150A MPPT solar charger
- BMS** Built-in BMS
-  LCD with Touch Pad
-  Parallel Operation up to 6 Units



NOVA 8.6KW
9.8KW

MODEL NAME	NOVA-8.6KW	NOVA-9.8KW
RATED POWER	8600VA/8600W	9800VA/9800W
Parallel Capability	Yes, 6 units	
INPUT		
Voltage	230VAC	
Selectable voltage range	170 ~ 280VAC (for computers) 90 ~ 280VAC (for home appliances)	
Frequency range	50 Hz / 60 Hz (auto sensing)	
OUTPUT		
AC voltage regulation (batt. mode)	230VAC ± 5%	
Surge power	16000VA	22000VA
Efficiency (peak)	93%	
Transfer time	10 ms (for personal computers) 20 ms (for home appliances)	
Waveform	Pure sine wave	
BATTERY		
Battery voltage	48VDC	
Floating charge voltage	54VDC	
Overcharge protection	63VDC	
SOLAR CHARGER & AC CHARGER		
Max. PV array power	14000W (7000W*2)	
MPPT range @ operating voltage	90 ~ 450VDC	
Max. PV array open circuit voltage	500VDC	
Max. PV input current	27A * 2 (MAX 40A)	
Max. solar charge current	150A	
Max. AC charge current	150A	
Max. charge current	150A	
GENERAL SPECIFICATION		
Dimension, D*W*H (mm)	158.4*503.6*530.8	147.4*432.5*553.6
Net weight (Kgs)	19.7	20
Communication interface	USB/RS232/RS485/Wi-fi/Dry-contact	
OPERATING ENVIRONMENT		
Humidity	5% to 95% relative humidity(non-condensing)	
Operating temperature	-10°C to 50°C	
Storage temperature	-15°C to 60°C	



PV - 14000

Single Phase Hybrid Inverter

NOVA 12.6KW

- 

Feed-in to the Grid
- 

Up to 15kW Gen/AC coupling input
- 

2 MPPT with 4 strings, Max. 24kW PV input
- 

Separate Generator Port Available
- 

Dual AC Output Port
- 

Max. charging and discharging current: 250A
- 

Integrated Design with Load and Battery Breaker
- 

Touch Screen
- 

Parallel Operations up to 16 Units



NOVA 12.6KW

MODEL NAME	NOVA-12.6KW
Rated Power	12600VA/12600W
INPUT (PV DC)	
Max. PV array power (W)	24000 (12000/12000)
Rated PV input voltage (V)	320
Number of independent MPPT inputs	2/(2:2)
PV input voltage range (V)	100-480
MPPT voltage range (V)	120-440
Start-up voltage (V)	100
Max. PV input current per MPPT (A)	35/35
Max. PV short-circuit current input per MPPT (A)	44/44
Battery	
Compatible battery type	Lithium-ion/Lead-Acid
Rated battery voltage (V)	48/51.2
Battery voltage range (V)	46.4-60/38.4-60
Max. charging / discharging current (A)	250
Max. charging / discharging power (W)	12600
Force wake up battery from PV and Grid function	YES
Grid	
Rated AC voltage (V)	230
Acceptable Input Voltage Range (V)	90~280
Rated AC frequency (Hz)	60/50
Rated AC input current (A)	100
Rated AC input power (W)	24000
PF	0.99
THDI	<5%
Max. continuous AC passthrough current(A)	100
UPS	
Rated output power (W)	12600
Rated output voltage (V)	230
Rated output current (A)	50
Rated output frequency (Hz)	50/60
Surge power, duration	2Pn, 5s
Switching time (UPS)	<7ms@Single; <10ms@Parallel
Wave form	Sine wave
THDV	3%
GEN	
Acceptable input voltage range	230
Max. AC input voltage (V)	280
Max. AC input current (A)	63
Rated input frequency	50Hz/60Hz (Auto detection)
Transfer time	<10ms@Single; <20ms@Parallel
THDV	<3%
Overload protection	5s@?150% load /10s@110%~150% load
Efficiency	
Max. efficiency	93.50%
Max. charging / discharging efficiency	93.00%
Max. MPPT efficiency	99.00%
Protection	
PV reverse polarity protection(Y/N)	YES
Over current / voltage protection(Y/N)	YES
AC Short-circuit current protection(Y/N)	YES
Grid monitoring(Y/N)	YES
DC switch(Y/N)	YES
DC / AC Surge protection Type III	Type III
Battery reverse polarity protection(Y/N)	YES
General	
Dimensions, D*W*H (mm)	150*530*830
Weight (Kgs)	40



PV - 24000

Single Phase Hybrid Inverter

ELEGO OYSTER 6.5KW 10KW



Feed-in to the Grid



Peak Shaving / AC Coupling Function



2 MPPT, Max. PV Input 6.5KW: 9.6KW
10KW: 15KW



Separate Generator Port Available
Intelligent Generator Control



IP66 Protection



Max. Charging & Discharging
Current: 210A

BMS

Built-in BMS



Smart Scheduling For Charge &
Discharge Cycles



Parallel Operation 6.5KW: 6 Units
up to: 10KW: 10 Units



6.5KW
10KW

MODEL NAME	ELEGO-OYSTER-6.5KW		ELEGO-OYSTER-10KW	
RATED POWER	6500VA/6500W		10000VA/10000W	
PV INPUT (DC)				
Max. PV array power (W)	9600		15000	
Rated PV input voltage (V)	360		360	
Number of independent MPPT inputs	2		2	
PV input voltage range (V)	100 ~ 530		100 ~ 600	
MPPT voltage range (V)	150 ~ 425		120 ~ 500	
Start-up voltage (V)	140		140	
Max. PV input current per MPPT (A)	18		25	
Max. PV short-circuit current input per MPPT (A)	32		34	
BATTERY				
Compatible battery type	Lithium / Lead-Acid		Lithium / Lead-Acid	
Rated battery voltage (V)	48		48	
Battery voltage range (V)	40 ~ 60		40 ~ 60	
Max. charge/discharge current (A)	125/145		208	
Max. charge/discharge power (W)	6500		10000	
Force wake up battery from PV function	Yes		Yes	
GRID				
Rated AC voltage (V)	230		230	
Rated AC frequency (Hz)	50/60		50/60	
Rated AC output current (A)	28.2		44	
Rated AC output power (W)	6500		10000	
Max. AC input power (W)	12000		15000	
Power factor			0.99	
THDI	<3%		<3%	
Max. continuous AC passthrough current (A)	52.2		90	
UPS				
Rated output power (W)	6500		10000	
Rated output voltage (V)	230		230	
Rated output current (A)	28.2		44	
Rated output frequency (Hz)	50/60		50/60	
Surge power, duration	2Pn, 0.5s		2Pn, 0.5s	
Switching time (UPS)	< 20ms		< 20ms	
Waveform	Sine wave		Sine wave	
THDV	< 5%		< 3%	
EFFICIENCY				
MPPT efficiency	99.9%		99.9%	
Max. efficiency	97.3%		97.5%	
Max. charge efficiency	94.5%		94.0%	
Max. discharge efficiency	94.5%		94.0%	
PROTECTION				
Over current / voltage protection	Yes		Yes	
Anti-islanding protection	Yes		Yes	
AC short-circuit current protection	Yes		Yes	
Leakage current protection	Yes		Yes	
Ground fault monitoring	Yes		Yes	
Grid monitoring	Yes		Yes	
DC switch	Yes		Yes	
DC surge protection type III	Yes		Yes	
AC surge protection type III	Yes		Yes	
GENERAL SPECIFICATION				
Dimensions, W*H*D (mm)	487*455*211.7		536*556*290	
Weight (Kgs)	29		42.64	
Ingress protection rating	IP66		IP66/NEMA 4X	
Operating environment temperature range (°C)	-25 ~ 60		0 ~ 100%	
Storage temperature range (°C)	-40 ~ 65		-25 ~ 60	
Relative humidity	0 ~ 95%		-25 ~ 60	
Display communication interface	LCD+RGB, RS485/wi-fi/CAN		Touch color screen, RS485/wi-fi/CAN	
Cooling method			Smart cooling	
Topology on AC side			Transformer-less	
Altitude (M)			<2000m	
Noise emission (typical)	<25dB		<50dB	



Three Phase Hybrid Inverter



5-20KW
(ThreePhase)



Supports Independent Grid Export Control for Each Phase



Unbalanced Load Support



3 MPPT with 6 Strings



Dual Output for Smart Load Management



IP66 Protection



Dedicated GEN Port for Auto GEN Control, Smart Load, AC Coupling



User-friendly Touch Color Screen



Parallel Operation Up to 10 Units



5-20KW
(ThreePhase)

MODEL NAME	EOI-OYS-5KW-3P	EOI-OYS-6KW-3P	EOI-OYS-8KW-3P	EOI-OYS-10KW-3P	EOI-OYS-12KW-3P	EOI-OYS-15KW-3P	EOI-OYS-20KW-3P
Input (PV DC)							
Max. PV input power (W)	7500	9000	12000	15000	18000	22500	30000
Rated PV input voltage (V)	690						
Number of independent MPPT inputs	3/(1:1:1)						
Max. PV input voltage (V)	1000						
MPPT voltage range (V)	200~900						
Start-up voltage (V)	100						
Max. PV input current per MPPT (A)	20/20/20						
Max. PV short-circuit current input per MPPT	25/25/25						
Battery							
Compatible battery type	Lithium-ion/Lead-Acid						
Battery voltage range (V)	40-60						
Max. charging/discharging current (A)	125	150	200	220	250	250	250
Max. charging/discharging power (W)	5000	6000	8000	10000	12000	12000	12000
Charging Strategy for Li-ion Battery	Self-adaption to BMS						
Input AC (Grid)							
Input voltage (3Φ)	3L/N/PE, 230/400Vac						
Rated AC frequency (Hz)	50/60						
Max. AC input power (W)	6250	7500	10000	12500	15000	18750	25000
Max. AC input current (A)	9	10.9	14.5	18.1	21.8	27.1	36.25
Output AC (Grid)							
Rated AC voltage (V)	3L/N/PE, 230/400Vac						
Rated AC frequency (Hz)	50/60						
Rated AC output power (W)	5000	6000	8000	10000	12000	15000	20000
Rated AC output current (A)	7.20	8.70	11.60	14.50	17.40	21.70	29.00
PF	0.99(Adjustable from 0.8 leading to 0.8 lagging)						
THDI	< 3%						
Max. continuous AC passthrough current (A)	30			50			70
GEN							
Rated GEN voltage (V)	3L/N/PE, 230/400Vac						
Rated GEN frequency (Hz)	50/60						
Rated GEN input current (A)	30						
Rated GEN input power (W)	20700						
UPS							
Rated output power (W)	5000	6000	8000	10000	12000	15000(PV+Battery)	20000(PV+Battery)
Rated output voltage (V)	3L/N/PE, 230/400Vac						
Rated output current (A)	7.20	8.70	11.60	14.50	17.40	21.70	29.00
Rated output frequency (Hz)	50/60						
Surge power, duration	1.5Pn, 5min						
Switching time	≤10ms						
Wave form	Sine wave						
THDV	<3%						
Efficiency							
Max. efficiency	97.70%						
Max. charge/discharge efficiency	94.0%						
Protection							
PV reverse polarity protection	YES						
Over current/voltage protection	YES						
Anti-islanding protection	YES						
AC Short-circuit protection	YES						
Leakage current protection	YES						
Grid monitoring	YES						
DC switch	YES						
Ingress protection	YES						
DC Surge protection	YES						
AC Surge protection	YES						
General							
Dimensions(W*H*D)	480*703*258mm / 18.9*27.7*10.2 in						
Weight	53kg / 116.8lbs						
Ingress protection rating	IP66						
Operating environment temperature range(°C)	-25~+60, >45 °C derating						
Storage temperature range(°C)	-25~+60						
Relative humidity	0~100						
Display & Communication interface	Touch color screen, RS485/Wifi/CAN						
Warranty	5/10 years						
Cooling method	Smart cooling						
Topology	Transformer-less						
Altitude(m)	<2000						
Noise emission(typical)	<50dB						



High Voltage Hybrid Inverter



6-30KW
(High Voltage)



6-30KW (High Voltage)



Supports Independent Grid Export Control for Each Phase



150% Unbalanced Load Support



Max. 3 MPPT & Max. PV input 45kW



Dual Output for Smart Load Management



IP66 Protection



Dedicated GEN Port for Auto GEN Control, Smart Load, AC Coupling



User-friendly Touch Color Screen



Parallel Operation Up to 10 Units

MODEL NAME	EOI-OYS-6KW-3PHV	EOI-OYS-8KW-3PHV	EOI-OYS-10KW-3PHV	EOI-OYS-12KW-3PHV	EOI-OYS-15KW-3PHV	EOI-OYS-20KW-3PHV	EOI-OYS-25KW-3PHV	EOI-OYS-30KW-3PHV
Input (PV DC)								
Max. PV input power (W)	9000	12000	15000	18000	22500	30000	37500	45000
Rated PV input voltage (V)				690				
Number of independent MPPT inputs	2/(1:1)			3/(2:2:2)				
Max. PV input voltage (V)				1000				
MPPT voltage range (V)				200~900				
Start-up voltage (V)				100				
Max. PV input current per MPPT (A)	20/20					40/40/40		
Max. PV short-circuit current input per MPPT	25/25 (A:1/B:1)					50/50/50 (A:2/B:2/C:2)		
Battery								
Compatible battery type				Lithium-ion/Lead-Acid				
Rated battery voltage (V)				400				
Battery voltage range (V)				100~800				
Max. charging voltage (V)				800				
Max. charging/discharging current (A)	50					75		
Max. charging/discharging power (W)	6000	8000	10000	12000	15000	20000	25000	30000
Force wake up battery from PV function				YES				
Grid								
Rated AC voltage (V)				3L/N/PE, 230/400Vac				
Rated AC frequency (Hz)				50/60				
Rated AC output current (A)	8.7	11.6	14.5	17.4	21.7	29	36.2	43.5
Rated AC output power (W)	6000	8000	10000	12000	15000	20000	25000	30000
Max. AC input current (A)	11.3	17.4	21.8	26.1	32.6	43.5	54.3	65.3
PF				0.99 (Adjustable from 0.8 leading to 0.8 lagging)				
THDI				<3%				
Max. continuous AC passthrough current (A)				75				
GEN								
Rated GEN voltage (V)				3L/N/PE, 230/400Vac				
Rated GEN frequency (Hz)				50/60				
Rated GEN input current (A)				30				
Rated GEN input power (W)				20700				
UPS								
Rated output power (W)	6000	8000	10000	12000	15000	20000	25000	30000
Rated output voltage (V)				3L/N/PE, 230/400Vac				
Rated output current (A)	8.7	11.6	14.5	17.4	21.7	29	36.2	43.5
Rated output frequency (Hz)				50/60				
Surge power, duration				1.5Pn, 5min				
Switching time				≤10ms				
Wave form				Sine wave				
THDV				<3%				
Efficiency								
Max. efficiency				97.4%				
Max. charge/discharge efficiency				97.2%				
Protection								
PV reverse polarity protection				YES				
Over current/voltage protection				YES				
Anti-islanding protection				YES				
AC Short-circuit protection				YES				
Leakage current protection				YES				
Grid monitoring				YES				
DC switch				YES				
Ingress protection				YES				
DC Surge protection				Type III				
AC Surge protection				Type III				
General								
Dimensions (W*H*D)				480*703*258mm / 18.9*27.7*10.2 in				
Weight	TBD			50kg / 110.2lbs				
Ingress protection rating				IP66				
Operating environment temperature range (°C)				-25~60				
Storage temperature range (°C)				-40~70				
Relative humidity				0~100%				
Display & Communication interface				Touch color screen, RS485/Wifi/CAN				
Warranty				5/10 years				
Cooling method	Natural cooling			Smart cooling				
Topology				Transformer-less				
Altitude				<2000				
Noise emission (typical)				<50dB				



Single Phase On-Grid Inverter



- On-Grid Single Phase MPPT Inverter
- IP66 Protection
- Max. PV Voltage: 600V
- AC Grid Frequency: 50Hz / 60Hz
- DC Switch, Anti-Islanding & Over Current Protection
- Type II & III DC / AC SPD
- High Efficiency: up to 97.6%
- Wi-Fi / 4G Plug Optional
- LCD With Touch Pad



MODEL NAME	VMGI-1KW-SP	VMGI-2KW-SP	VMGI-3KW-SP
RATED POWER	1000VA/1000W	2000VA/2000W	3000VA/3000W
INPUT (DC)			
Max. DC voltage	600VDC		
Nominal voltage	380VDC		
Start voltage	60V	80V	80V
MPPT voltage range	60V ~ 560V	80V ~ 560V	80V ~ 560V
Number of MPPT	1		
Strings per MPPT	1		
Max. input current per MPPT	13A		
Max. short-circuit current per MPPT	15.6A		
OUTPUT (AC)			
Nominal AC output power	1000W	2000W	3000W
Max. AC apparent power	1100VA	2200VA	3300VA
Nominal AC voltage	230V L-N		
AC grid frequency range	50Hz / 60Hz ± 5Hz		
Max. output current (A)	4.8A	9.6A	14.4A
Power factor	0.8 leading - 0.8 lagging		
THDI	< 3%		
EFFICIENCY			
Max. efficiency	97.00%	97.50%	97.60%
Euro efficiency	96.50%	97.00%	97.00%
PROTECTION DEVICES			
DC switch	Yes		
Anti-islanding protection	Yes		
Output over current protection	Yes		
DC reverse polarity protection	Yes		
DC / AC surge protection	DC type II-III ; AC type II-III		
Insulation detection	Yes		
AC short circuit protection	Yes		
GENERAL SPECIFICATION			
Dimensions, W*H*D (mm)	350*290*120		
Weight (Kgs)	7.3	8	8
ENVIRONMENT			
Operating temperature range	-25°C ~ +60°C		
Cooling type	Natural convection		
Max. operating altitude	4000 m		
Max. operating humidity	0 - 100%		
AC output terminal type	Quick connector		
IP class	IP66		
Topology	Transformer-less		
Communication interface	RS-485 / Wi-fi / 4G		
Display	LCD / Bluetooth + App		

Single Phase On-Grid Inverter

VOL

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Ultimate Technology Provider

3.6KW-8KW





On-Grid Single Phase MPPT Inverter



IP66 Protection



Max. PV Voltage: 600V



AC Grid Frequency: 50Hz / 60Hz



DC Switch, Anti-Islanding & Over Current Protection



Type II & III DC / AC SPD



High Efficiency: Up To 98.3%



Wi-Fi / 4G Plug Optional



LCD With Touch Pad

MODEL NAME	VMGI-3.6KW-SP	VMGI-4KW-SP	VMGI-5KW-SP	VMGI-6KW-SP	VMGI-8KW-SP
RATED POWER	3600VA/3600W	4000VA/4000W	5000VA/5000W	6000VA/6000W	8000VA/8000W
INPUT (DC)					
Max. DC voltage	600V				
Nominal voltage	380V				
Start voltage	120V	120V	120V	120V	100V
MPPT voltage range	80V ~ 560V	80 V ~ 560V	80V ~ 560V	80V ~ 560V	80V ~ 540V
Number of MPPT	2				
Strings per MPPT	1				
Max. input current per MPPT	15A	15A	15A	15A	26A / 16A
Max. short-circuit current Per MPPT	18A	18A	18A	18A	31A/ 19A
OUTPUT (AC)					
Nominal AC output power	3600W	4000W	5000W	6000W	8000W
Max. AC apparent power	3960VA	4400VA	5500VA	6000VA	8000VA
Nominal AC voltage	230V L-N				
AC grid frequency range	50Hz / 60Hz ± 5Hz				
Max. output current	17A	19A	24A	26A	35A
Power factor	0.8 leading - 0.8 lagging				
THDI	< 3%				
EFFICIENCY					
Max. efficiency	98.1%	98.3%	98.3%	98.3%	98.3%
Euro efficiency	97.7%	97.9%	97.9%	97.9%	97.9%
PROTECTION DEVICES					
DC switch	Yes				
Anti-islanding protection	Yes				
Output over current protection	Yes				
DC reverse polarity protection	Yes				
DC / AC surge protection	DC type II-III ; AC type II III				
Insulation detection	Yes				
AC short circuit protection	Yes				
GENERAL SPECIFICATION					
Dimensions W*H*D (mm)	380*380*150				
Weight (Kgs)	10	11	11	11	13
Operating temperature range	-25°C ~ +60°C				
Cooling type	Natural convection			Fan cooling	
Max. operating altitude	4000 m				
Max. operating humidity	100%				
AC output terminal type	Quick connector				
IP class	IP65				
Topology	Transformer-less				
Communication	RS-485 / Wi-fi / 4G				
Display	LCD / Bluetooth + App				



Single Phase On-Grid Inverter





Ongrid Single Phase MPPT Inverter



IP66 Protection



Max. PV voltage: 600V



AC Grid Frequency: 50Hz / 60Hz



DC Switch, Anti-Islanding & Over Current Protection



Type II & III DC / AC SPD



High Efficiency: Up to 98.1%



Wi-fi / 4G Plug Optional



LCD With Touch Pad

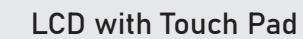
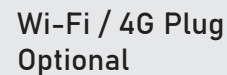
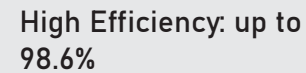
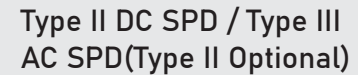
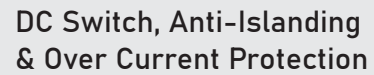
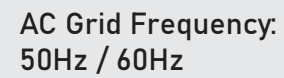
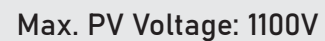
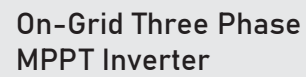


MODEL NAME	VMGI-10KW-SP
Max. DC power	15000W
Max. DC voltage	600Vd.c.
Normal DC voltage	360Vd.c.
Start Voltage	80Vd.c.
MPPT voltage range	60-550Vd.c.
Number of MPPT trackers	3
Strings per MPPT tracker	1
Max. input current per MPPT tracker	15Ad.c. *3
PV short circuit current	22.5Ad.c.*3
Normal AC output Power	10000W
Max. AC output Power	10000W
Max. Apparent Power	10000VA
Normal AC voltage	220/230Va.c.
Normal AC grid frequency	50/60Hz
Max. output current	45.5Aa.c.
Inrush current	< 45.5Aa.c.
Max. output fault current	102Ad.c.
Max. inverter backfeed current to the array	0Ad.c.
Max. output overcurrent protection	75Ad.c.
Output Power Factor	~1 (Adjustable from 0.8 leading to 0.8 lagging)
THDI	<3%
Topology	Transformer less
Max. efficiency	98.1%
Euro-eta	97.5%
Dimensions (W / L / D)	380*483*161mm
Weight(KG)	15
Operating temperature range	-25 °C~+60 °C
Relative humidity	0-100%
Operating Altitude	≤4000m
Noise (typical)	≤ 35dB
Cooling concept	Natural
Protection rating	IP66
Protection class	I
Overvoltage category (PV/AC)	II/III
Pollution degree	2
User Interface	LED & LCD
Communication	RS485/WIFI/GPRS

VOL **MORE** **3KW-12KW**
Ultimate Technology Provider



3KW-1 2KW



**10
YEARS
WARRANTY**

WARRANTY
Terms & Conditions Apply

MODEL NAME	VMGI-3KW	VMGI-4KW	VMGI-5KW	VMGI-6KW	VMGI-8KW	VMGI-10KW	VMGI-12KW
RATED POWER	3000VA/3000W	4000VA/4000W	5000VA/5000W	6000VA/6000W	8000VA/8000W	10000VA/10000W	12000VA/12000W
INPUT (DC)							
Max. DC voltage	1100V						
Nominal voltage	650V						
Start voltage	250V						
MPPT voltage range	140V ~ 1000V						
Number of MPPT	2						
Strings per MPPT	1						
Max. input current per MPPT	15A						
Max. short-circuit current per MPPT	20A						
OUTPUT (AC)							
Nominal AC output power	3000W	4000W	5000W	6000W	8000W	10000W	12000W
Max. AC output power	3300VA	4400VA	5500VA	6600VA	8800VA	11000VA	13200VA
Nominal AC voltage	400 / 230 V, 3P + N - PPE						
AC grid frequency range	50 / 60 Hz (± 5Hz)						
Max. output current	4.8A	6.4A	8.0A	9.6A	12.8A	16.0A	19.2A
Power factor	0.8 leading - 0.8 lagging						
THDI	3%						
EFFICIENCY							
Max. efficiency	98.4%	98.4%	98.4%	98.4%	98.6%	98.6%	98.6%
Euro efficiency	97.5%	97.5%	97.5%	97.5%	98.0%	98.1%	98.1%
PROTECTION DEVICES							
DC switch	Yes						
Output over current protection	Yes						
Anti-islanding protection	Yes						
DC reverse polarity protection	Yes						
String fault detection	Yes						
DC / AC surge protection	DC type II; AC type III; type II optional						
Insulation detection	Yes						
AC short circuit protection	Yes						
GENERAL SPECIFICATION							
Dimensions, W*H*D (mm)	380*483*161						
Display	LCD						
Operating temperature range	-25°C ~ +60°C						
Cooling type	Natural cooling						
Max. operating altitude	4000 m						
Max. operating humidity	0 - 100% (non-condensing)						
AC output terminal type	Connector						
IP class	IP66						
Topology	Transformerless						
Communication	RS-485 / Wi-fi / 4G						

Three Phase On-Grid Inverter



 On-Grid Three phase MPPT Inverter	 IP66 Protection	 Max. PV Voltage: 1100V
 AC Grid Frequency: 50Hz / 60Hz	 DC Switch, Anti-Islanding & Over Current Protection	 Type II DC SPD / Type III AC SPD (Type II Optional)
 High Efficiency: Up to 98.6%	 Wi-Fi / 4G Plug Optional	 LCD With Touch Pad



MODEL NAME	VMGI-15KW	VMGI-20KW	VMGI-25KW
RATED POWER	15000VA/15000W	20000VA/20000W	25000VA/25000W
INPUT(DC)			
Max. DC voltage	1100V	1100V	1100V
Nominal voltage	650V	650V	650V
Start voltage	250V	250V	250V
MPPT voltage range	140V ~ 1000V	140V ~ 1000V	140V ~ 1000V
Number of MPPT tracker	2	2	2
Strings per MPPT tracker	2/1	2	2
Max. input current per MPPT	30A/15A	30A	30A
Max. short-circuit current per MPPT	40A/20A	40A	40A
OUTPUT(AC)			
Nominal AC output power	15000W	20000W	25000W
Max. AC output power	16500VA	22000VA	27500VA
Nominal AC voltage	400V 3L+ N		
AC grid frequency range	50Hz / 60Hz		
Max. output current	23.9A	31.9A	39.9A
Power factor	0.8leadlng - 0.8lagging		
THDI	3%		
EFFICIENCY			
Max. efficiency	98.6%	98.6%	98.6%
Euro efficiency	98.2%	98.3%	98.3%
PROTECTION DEVICES			
DC switch	Yes		
Output over current	Yes		
Anti-islanding protection	Yes		
DC reverse polarity protection	Yes		
String fault detection	Yes		
AC/DC surge protection	DC: type II / AC: type III / type II optional		
Insulation detection	Yes		
AC short circuit protection	Yes		
GENERAL SPECIFICATION			
Dimensions, W*H*D (mm)	380*483*193		
Weight (Kgs)	20.7		
Operating temperature range	-25°C ~ +60°C		
Cooling type	Fan cooling		
Max. operation altitude	4000m		
Max. operation humidity	0-100% (non-condensing)		
AC output terminal type	Connector		
IP class	IP66		
Topology	Transformer-less		
Communication	RS485/Wi-fi/4G		
Display	LCD		

Three Phase On-Grid Inverter



- On-Grid Three phase MPPT Inverter
- IP66 Protection
- Max. PV Voltage: 1100V
- AC Grid Frequency: 50Hz / 60Hz
- DC Switch, Anti-Islanding & Over Current Protection
- Type II DC SPD / Type III AC SPD (Type II Optional)
- High Efficiency: Up to 98.6%
- Wi-fi / 4G Plug Optional
- LCD With Touch Pad



MODEL NAME	VMGI-30KW	VMGI-40KW
RATED POWER	30000VA/30000W	40000VA/40000W
INPUT(DC)		
Recommended Max. PV array input power @ STC	39KW	52KW
Max. DC voltage	1100V	
Nominal voltage	650V	
Start voltage	250V	
MPPT voltage range	200V ~ 1000V	
Number of MPPT tracker	3	
Strings per MPPT tracker	2	
Max. input current per MPPT	30A	
Max. short-circuit current per MPPT	40A	
OUTPUT (AC)		
Nominal AC output power	30000W	40000W
Max. AC output power	33000VA	44000VA
Nominal AC voltage	230/400V, 3P+N+PE	
AC grid frequency range	50 / 60Hz (±5Hz)	
Max. output current	47.8A	63.8A
Power factor	0.8leading - 0.8lagging	
THDI	3%	
EFFICIENCY		
Max. efficiency	98.7%	
Euro efficiency	98.4%	
PROTECTION DEVICES		
DC switch	Yes	
Output over current	Yes	
Anti-islanding protection	Yes	
DC reverse polarity protection	Yes	
String fault detection	Optional	
DC/AC surge protection	DC: type II/AC: type III / II optional	
Insulation detection	Yes	
AC short circuit protection	Yes	
GENERAL SPECIFICATION		
Dimensions, W*H*D (mm)	380*483*223	380*483*227
Weight (Kgs)	25.5	32.5
Operating temperature range	- 25°C ~ +60°C	
Cooling type	Fan cooling	
Max. operation altitude	4000m	
Max. operation humidity	0-100% (non-condensing)	
AC output terminal type	Connector	
IP class	IP66	
Topology	Transformerless	
Communication	RS485/Wi-fi/4G	
Display	LCD	

Three Phase On-Grid Inverter



On-Grid Three phase MPPT Inverter

IP66 Protection

Max. PV Voltage: 1100V

AC Grid Frequency: 50Hz / 60Hz

DC Switch, Anti-Islanding & Over Current Protection

Type II DC SPD / Type III AC SPD (Type II Optional)

High Efficiency: Up to 98.6%

Wi-Fi / 4G Plug Optional Reactive Power Control

LCD With Touch Pad



MODEL NAME	VMGI-50KW	VMGI-60KW	VMGI-70KW	VMGI-80KW
RATED POWER	50000VA/50000W	60000VA/60000W	70000VA/70000W	80000VA/80000W
INPUT (DC)				
Max. DC voltage	1100V	1100V	1100V	1100V
Nominal voltage	650V	650V	650V	650V
Start voltage	250V	250V	250V	250V
MPPT voltage range	200 V~1000V	200 V~1000V	200 V~1000V	200 V~1000V
Number of MPPT	4	4	4	4
Strings per MPPT	2	2	3	3
Max. input current per MPPT	32A	32A	45A	45A
Max. short-circuit current per MPPT	48A	48A	60A	60A
OUTPUT(AC)				
Nominal AC output power	50000W	60000W	70000W	80000W
Max. AC output power	55000VA	66000VA	77000VA	88000VA
Nominal AC voltage	400V 3L+N+PE			
AC grid frequency range	50Hz / 60Hz (±5Hz)			
Max. output current	79.7A	95.6A	111.6A	127.5A
Power factor	0.8leading - 0.8lagging			
THDI	3%			
EFFICIENCY				
Max. efficiency	98.5%	98.5%	98.6%	98.6%
Euro efficiency	98.2%	98.2%	98.3%	98.3%
PROTECTION DEVICES				
DC switch	Yes			
Output over current protection	Yes			
Anti-islanding protection	Yes			
DC reverse polarity protection	Yes			
String fault detection	Yes			
DC/AC surge protection	Type II			
Residual current monitoring	Yes			
AC short circuit protection	Yes			
GENERAL SPECIFICATION				
Dimensions, W*H*D (mm)	515*585*287			
Weight (Kgs)	51	55		
Operating temperature range	-25℃ ~ +60℃			
Cooling type	Fan cooling			
Max. operating altitude	4000m			
Max. operating humidity	0-100%(no condensation)			
AC output terminal type	OT terminal			
IP class	IP66			
Topology	Transformer-less			
Communication	RS485/Wi-fi/4G			
Display	LCD			

Three Phase On-Grid Inverter



On-Grid Three phase MPPT Inverter

IP66 Protection

Max. PV Voltage: 1100V

AC Grid Frequency: 50Hz / 60Hz

DC Switch, Anti-Islanding & Over Current Protection

Type II DC SPD / Type II AC SPD

High Efficiency: Up to 98.7%

Wi-Fi / 4G Plug Optional Reactive Power Control

LCD With Touch Pad



MODEL NAME		VMGI-125KW
INPUT (DC)		
Max. DC voltage		1100V
Max. input current per MPPT		40A
Max. short-circuit current per MPPT		60A
Start voltage		350V
MPPT voltage range		200 ~ 1000V
Nominal voltage		650V
Number of MPPT		8
Strings per MPPT		2
OUTPUT (AC)		
Nominal AC output power		125KW
Max. AC apparent power		125KVA
Nominal AC voltage		230V/400V, 3W+ PE, 3W+ N+PE
AC grid frequency range		50Hz / 60Hz (±5Hz)
Max. output current		181.2A
Power factor		0.8 leading - 0.8 lagging
THDI		< 3% (nominal power)
EFFICIENCY		
Max. efficiency		98.7%
Euro efficiency		98.5%
PROTECTION DEVICES		
DC switch		Yes
Anti-islanding protection		Yes
Output over current protection		Yes
DC reverse polarity protection		Yes
String fault detection		Optional
DC/AC surge protection		DC type II; AC type II
AC short circuit protection		Yes
AFCI function		Optional
Night SVG function		Optional
PID recovery		Optional
Insulation detection		Yes
Residual current monitoring		Yes
GENERAL SPECIFICATION		
Dimensions, W*H*D (mm)		965*700*355
Weight (Kgs)		85
Operating temperature range		-30 ~ 60°C
Cooling type		Fan Cooling
Max. operation altitude		5,000 m (> 4,000m derating)
Max. operating humidity		0-100%
Ipclass		IP66
Noise (dB)		75dB
Topology		Transformer-less
Communication		RS485/PLC/Wi-fi/4G
Display		LED , Bluetooth + APP

Three Phase On-Grid Inverter



- On-Grid Three phase MPPT Inverter
- IP66 Protection
- Max. PV Voltage: 1100V
- AC Grid Frequency: 50Hz / 60Hz
- DC Switch, Anti-Islanding & Over Current Protection
- Type II DC / AC SPD Smart I-V Curve
- High Efficiency: Up to 99%
- Wi-Fi / 4G Plug Optional
- LCD With Touch Pad



MODEL NAME	VMGI-350KW-1	VMG-350KW-2
INPUT (DC)		
Max. DC voltage	1500V	
Max. input current per MPPT	30A	40A
Max. short-circuit current per MPPT	45A	60A
Start voltage	500V	
MPPT voltage range	500V ~ 1500V	
Number of strings	32	24
Number of MPPT	16	12
Strings per MPPT	2	
OUTPUT (AC)		
Nominal AC output power	350KW	
Max. AC apparent power	352KVA	
Nominal AC voltage	800V, 3W+ PE	
Nominal frequency	50Hz / 60Hz (±5Hz)	
Frequency range	45-55Hz / 55-65Hz	
Max. output current	254A	
Power factor	0.8leading - 0.8lagging	
THDI	< 3% (nominal power)	
EFFICIENCY		
Max. efficiency	99%	
PROTECTION DEVICES		
DC switch	Yes	
Anti-islanding protection	Yes	
Over current protection	Yes	
DC reverse polarity protection	Yes	
String fault detection	Yes	
DC/AC surge protection	Type II	
AC short circuit protection	Yes	
Residual current detection	Yes	
PID recovery	Optional	
Night SVG function	Optional	
GENERAL SPECIFICATION		
Dimensions, W*H*D (mm)	1050*860*393	
Weight (Kgs)	130	
Operating temperature range	-30 ~ 60°C	
Cooling type	Fan cooling	
Max. operation altitude	5,000 m (> 4,000m derating)	
Max. operating humidity	0 -100%	
AC output terminal type	OT terminal	
IP class	IP66	
Noise (dB)	75dB	
Topology	Transformer-less	
Communication	RS485 / PLC	
Display	LED , Buletooth + APP	

Solar Water Pump Controller

CM-SPC50 2.2-37KW



CM-SPC50 2.2-37KW

- 

Water Pump Protection
- 

Web/App Monitoring,
Touch Screen
- 

Stable Operation Under
Weak Light
- 

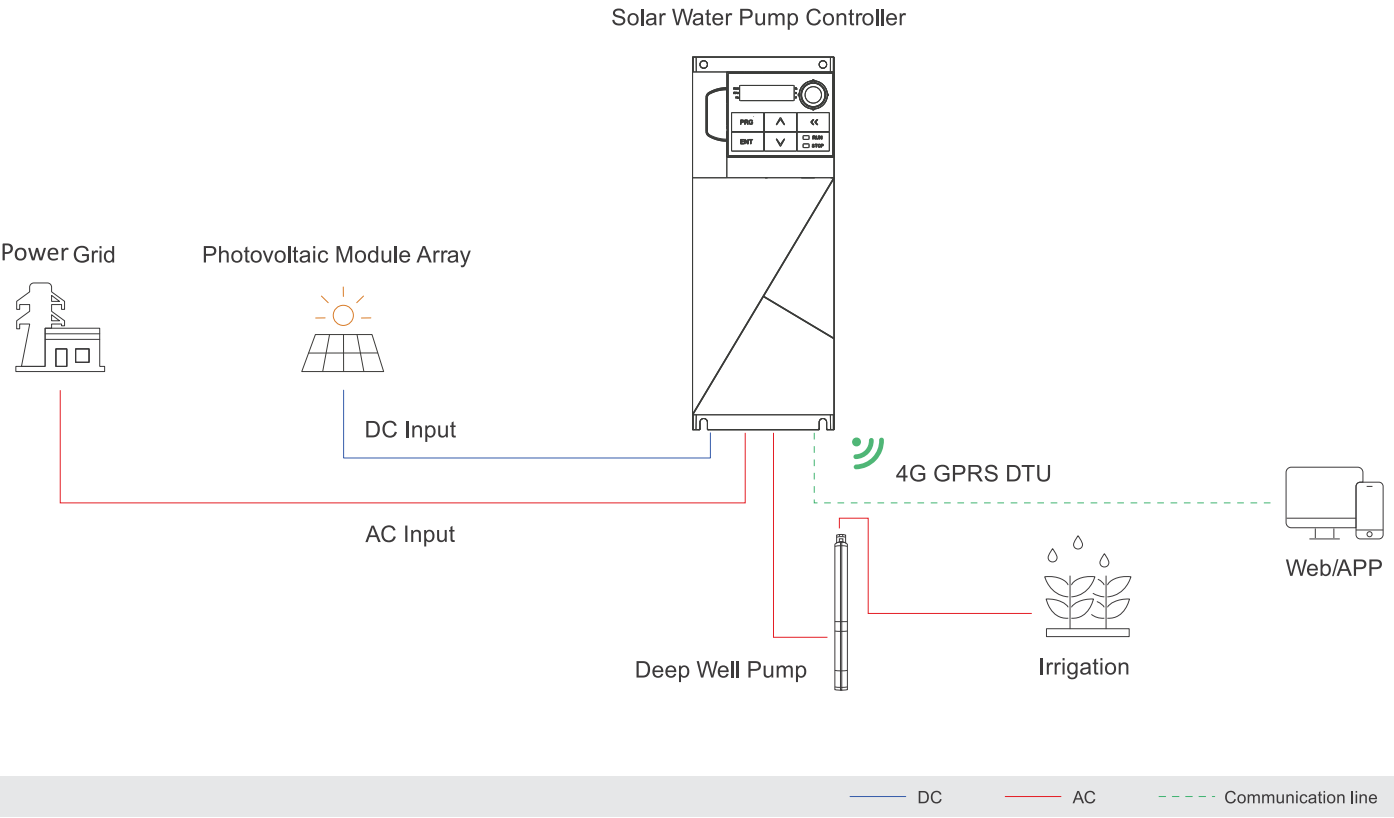
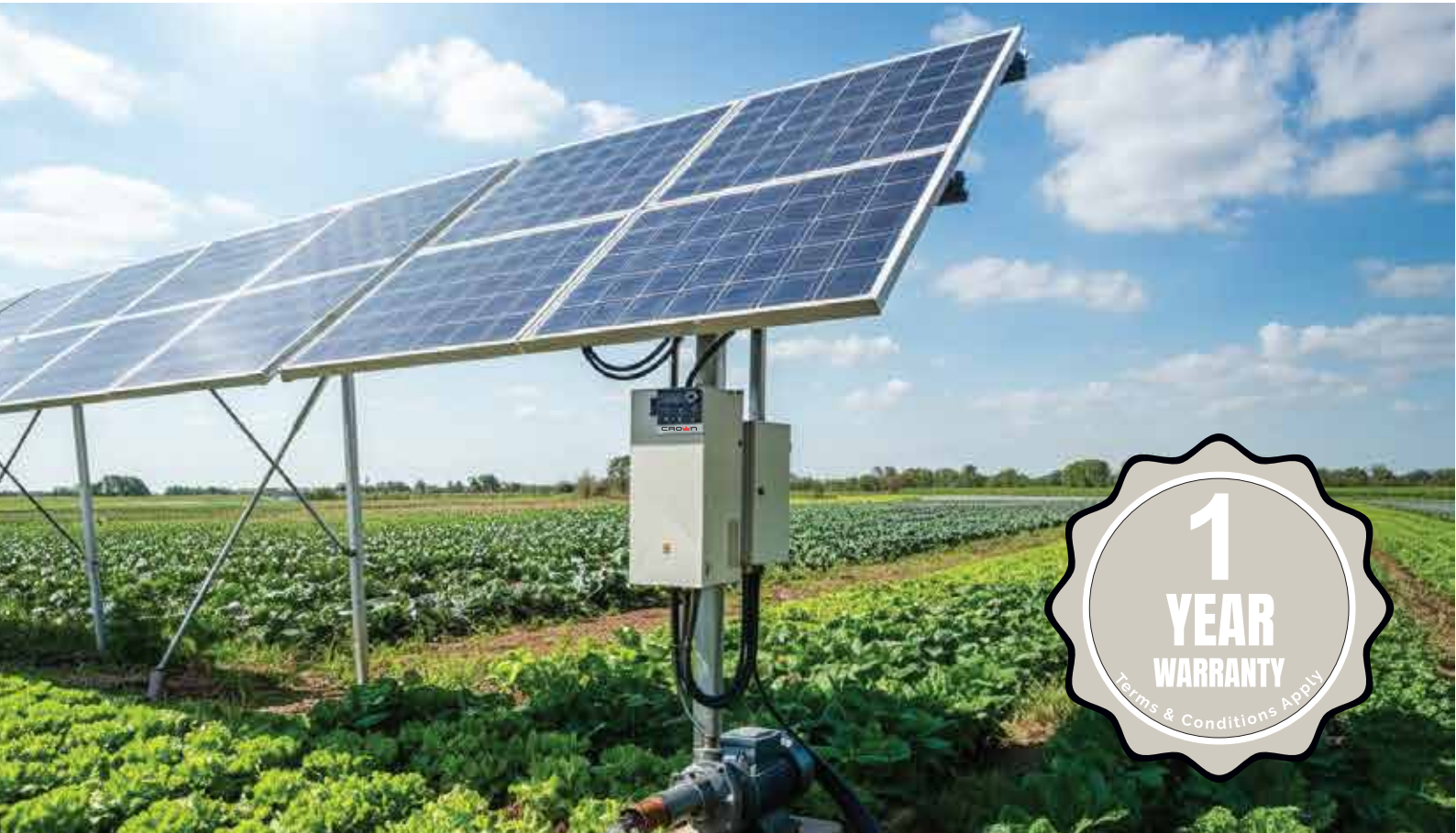
220-900 VDC MPPT
Input Range
- 

AC & DC Power Support
- 

Auto-Switches Between
Solar and Grid Power
- 

Precise Speed & Torque,
High Efficiency

MODEL NAME	Rated Power (kW)	Power Supply Capacity (kVA)	Input Current (A)	Output Current (A)	Adapted Motor	
					kW	HP
150-450VDC / Single phase 200-240VAC input Three phase 0-220VAC output						
CM-SPC50-2.2KWSP	2.2	4	23	9.6	2.2	3
220-900VDC / Three phase 380-480VAC input Three phase 0-380VAC output						
CM-SPC50-2.2KW	2.2	4	5.8	5.1	2.2	3
CM-SPC50-4KW	4	5.9	10.5	9	4	5.5
CM-SPC50-5.5KW	5.5	8.9	14.6	13	5.5	7.5
CM-SPC50-7.5KW	7.5	11	20.5	17	7.5	10
CM-SPC50-11KW	11	17	26	25	11	15
CM-SPC50-15KW	15	21	35	32	15	20
CM-SPC50-18.5KW	18.5	24	38.5	37	18.5	25
CM-SPC50-22KW	22	30	46.5	45	22	30
CM-SPC50-30KW	30	40	62	60	30	40
CM-SPC50-37KW	37	57	76	75	37	50



Solar Water Pump Controller

CM-SPC33
45-110KW



CM-SPC33 45-110KW



Up to 99.8% MPPT
Efficiency



Multi Pump Protection



AM PMSM Single Phase,
Multi Types of Pumps



Automatic Hybrid
Compensation



Automatic Start & Stop



Zero Programming



MODEL NAME	CM-SPC 33-45KW	CM-SPC 33-55KW	CM-SPC 33-75KW	CM-SPC 33-93KW	CM-SPC 33-110KW
DC Input					
Max. Input Voltage [V]	900				
Start Voltage [V]	350				
Min. Input Voltage [V]	220				
MPPT Voltage Range [V]	500~750				
Recommended Input Working Voltage [V]	560				
Recommended PV Array Power [kW]	54~72	66~88	90~120	112~149	132~176
AC Input					
Input Voltage [V]	3PH 380-480				
Input Frequency [Hz]	50/60				
AC Output					
Output Power [kW]	45	55	75	93	110
Rated Current [A]	91	112	150	176	210
Output Voltage [V]	3PH 0-380/480				
Output Frequency [Hz]	0-50/60				
Control Performance					
Control Mode	V/F , SVC				
Overload Capability	150% of rated current:60 seconds;180% of rated current:1 seconds				
Protection					
Undervoltage Protection	Integrated				
Overvoltage Protection	Integrated				
Overload Protection	Integrated				
Short-circuit Protection	Integrated				
Overheat Protection	Integrated				
Lack of Water Protection	Integrated				
Surge Protection	Integrated				
General Data					
MPPT Efficiency	99.8%				
Ingress Protection Degree	IP20				
Cooling Mode	Air cooling				
Communication Interface	RS485 communication interface				
Ambient Operating Temperature	-20°C ~ 60°C (>45°C, rated power will be derated for 1% by every additional 1°C)				
Operation Altitude	Below 2000 meters, there is no need to derate, 2000-3000 meters, each increase of 100 meters, derate 1%				
Physical Dimension [H*W*D] [mm]	543*310*280		580*358*328		
Weight [kg]	22.3	22.7	38.93	39.45	40.07
Standard Warranty [Month]	18				

Notes: Inductor is suggested to be added to the solar pumping system if the distance between solar pump controller and water pump is over 50m

Solar Water Pump Controller

CM-SPC33
132-200KW



CM-SPC33 132-200KW



Up to 99.8% MPPT Efficiency



Multi Pump Protection



AM PMSM Single Phase, Multi Types of Pumps



Automatic Hybrid Compensation



Automatic Start & Stop



Zero Programming

MODEL NAME	CM-SPC 33-132KW		CM-SPC 33-160KW	CM-SPC 33-200KW
DC Input				
Max. Input Voltage [V]	900			
Start Voltage [V]	350			
Min. Input Voltage [V]	220			
MPPT Voltage Range [V]	500~750			
Recommended Input Working Voltage [V]	560			
Recommended PV Array Power [kW]	158~211	192~256	240~320	
AC Input				
Input Voltage [V]	3PH 380-480			
Input Frequency [Hz]	50/60			
AC Output				
Output Power [kW]	132	160	200	
Rated Current [A]	253	304	377	
Output Voltage [V]	3PH 0-380/480			
Output Frequency [Hz]	0-50/60			
Control Performance				
Control Mode	V/F, SVC			
Overload Capability	150% of rated current:60 seconds;180% of rated current:1 seconds			
Protection				
Undervoltage Protection	Integrated			
Overvoltage Protection	Integrated			
Overload Protection	Integrated			
Short-circuit Protection	Integrated			
Overheat Protection	Integrated			
Lack of Water Protection	Integrated			
Surge Protection	Integrated			
General Data				
MPPT Efficiency	99.8%			
Ingress Protection Degree	IP20			
Cooling Mode	Air cooling			
Communication Interface	RS485 communication interface			
Ambient Operating Temperature	-20°C ~ 60°C(>45°C, rated power will be derated for 1% by every additional 1°C)			
Operation Altitude	Below 2000 meters, there is no need to derate, 2000-3000 meters, each increase of 100 meters, derate 1%			
Physical Dimension [H*W*D] [mm]	1199*502*355	1570*600*408	1696*800*246.3	
Weight [kg]	87.68	88.42	179.8	
Standard Warranty [Month]	18			

Note: Inductor is suggested to be added to the solar pumping system if the distance between solar pump controller and water pump is over 50m.



All-In-One ESS



-  Feed-in to the Grid
-  ESS 6.5KW Inverter & 5kWh LiFeP04 Battery
-  Dual MPPT with 17A Each with Max. PV Input Power 9600W
-  ON / OFF Grid Switching
-  Expandable Up to 15kWh
-  Dual Output for Smart Load Management
-  PV or AC or Generator Wake Up Battery
-  User-friendly Touch Screen
-  Parallel Operation Up to 16 Units



MODEL NAME	NURA6.5kW
INPUT (PV DC)	
Max. PV input power (W)	9600 (4800/4800)
Rated PV input voltage (V)	320
Number of independent MPPT inputs	2 / (1:1)
PV input voltage range (V)	100 ~ 500
MPPT voltage range (V)	120 ~ 400
Start-up voltage (V)	100
Max. PV input current per MPPT (A)	17 / 17
Max. PV short-circuit current input per MPPT (A)	25 / 25
Max. PV charging current (A)	140
Battery Input	
Battery voltage (V)	38.4 ~ 60
Max. charging /discharging current (A)	140
Max. charging /discharging power (W)	6500
UPS output @ off-grid	
Output voltage waveform	Pure sine wave
Output voltage regulation (V)	208 / 220 / 230 / 240±5%
Output frequency (Hz)	50 / 60
Rated output power (W)	6500
Rated output current (A)	28.3
Peak efficiency	93%
Overload protection	5s@≥150% overload; 10s@110%~150% overload
Surge power	2*Pn, 5s
Parallel quantity	16pcs
AC Input	
Nominal input voltage (V)	230
Nominal input frequency (Hz)	50 / 60 (Auto detection)
Max. AC input current (A)	39.5
Max. AC input power (W)	9000
Max. AC charging current (A)	140
AC Ouput @ Gird	
Rated AC output current (A)	28.3
Rated AC output power (W)	6500
Rated AC current of bypass relays (A)	60
Switching time	Typical<7ms@Single; <20ms@Parallel
Gen	
Rated GEN voltage (V)	220 / 230 / 240
Rated GEN frequency (Hz)	50 / 60
Rated GEN input Current (A)	32
Rated GEN input Power (W)	7370
Rated GEN current of bypass relays (A)	60
Battery MODULE	
Battery power (kwh)	5.12
Rated voltage (V)	51.2
Rated capacity (Ah)	100
Battery type	Lithium-ion
Max. parellel (Default)	2PCS
Max. discharging current (A)	100
Max. charging current (A)	100
Dimension, W*H*D (mm)	580*434*165
Net weight (kgs)	54
Standard	
Protection / General Specifications	
Grid monitoring	YES
AC surge protection	Type III
Dimension, W*H*D (mm)	580*1300*165mm
Weight (kgs)	125.4
Ingress protection rating	IP21
Display & communication interface	Touch screen, RS485 / Wi-fi / CAN
Cooling method	Smart cooling
Topology	Transformer-less
Noise emission (typical) (dB)	<58
Operating temperature range (°C)	0 ~ 45
Storage temperature (°C)	-15 ~ 60
Humidity	5% to 95% Relative Humidity (Non-condensing)
Altitude (m)	<2000

Transparent LiFePO4 Battery



CMLB-12.8V-100 CMLB-25.6V-100
CMLB-12.8V-200 CMLB-25.6V-200



Eco-friendly Transparent ABS Casing



Grade A Prismatic LiFePO4 Cells



10 Years Designed Service Life

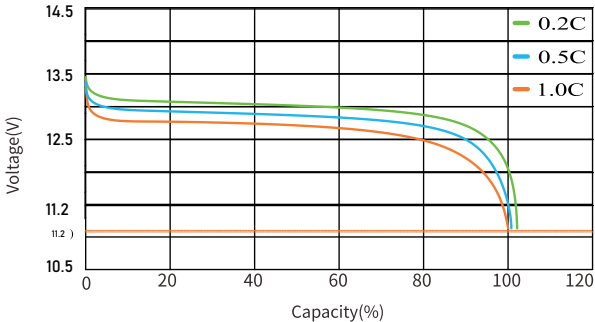


>4000 Cycles @ 80% DoD, 0.5C @ 25°C

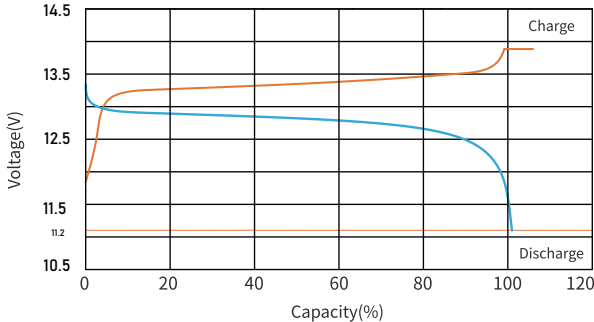


IP65 Protection

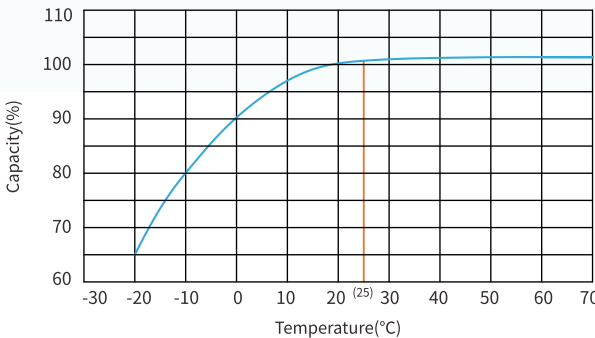
Discharge Performance at 25°C



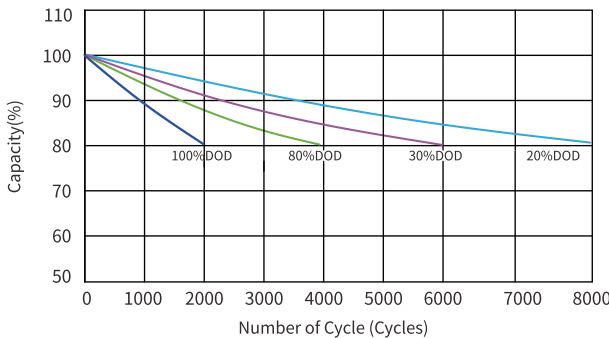
Charge and Discharge at 25°C, 0.5C



Temperature Effects on Capacity at 0.5C



Cycle Life with DOD @ 0.5°C, 25°C



MODEL NAME	CMLB-12.8V-100	CMLB-12.8V-200	CMLB-25.6V-100	CMLB-25.6V-200
Basic Performance				
Nominal Voltage (VDC)	12.8	12.8	25.6	25.6
Nominal Capacity (Ah)	100	200	100	200
Energy (kWh)	1.28	2.56	2.56	5.12
Internal Resistance (mΩ)	≤15	≤10	≤15	≤10
System Series/Parallel	4S, 4P	4S, 4P	2S, 4P	2S, 4P
Self Discharge	≤2%/Month			
Cycle Life	>6000 times @ 95% DOD			
Design Life	10 Years			
Charge Performance				
Recommended Charge Current (A)	50	100	50	200
Maximum Charge Current (A)	100	200	100	200
Recommended Charge Voltage (V)	13.6~13.8	13.6~13.8	13.6~13.8	13.6~13.8
Max Charge Voltage (V)	14.4	14.4	14.4	14.4
Overcharge Protection Voltage (V)	14.8~15.0	14.8~15.0	14.8~15.0	14.8~15.0
Charge Cut-off Voltage	>14.8V (0.5~1.5s)	>14.8V (0.5~1.5s)	>14.8V (0.5~1.5s)	>14.8V (0.5~1.5s)
Charge Mode	CC/CV			
Discharge Performance				
Maximum Continuous Discharge Currer	100	200	100	200
Pulse Current (≤5s)	600	600	600	750
Rated Discharge Cut-off Voltage (V)	11.2	11.2	22.4	20.8
Minimum Discharge Cut-off Voltage (V)	>10	>10	>20	>20
Reconnect Voltage (V)	>10.4	>10.4	>20.8	>20.8
Short Circuit Protection (μs)	200~800			
Mechanical Specifications				
Weight (kg)	10.1	19.5	21	37
Size (mm)	306*169*215	520*240*220	520*240*220	520*268*220
Case Material	ABS			
Terminal Type	M8			
IP Rating	IP65			
Temperature Performance				
Discharge Temperature (°C)	-20~60			
Charge Temperature (°C)	0~45			
Storage Temperature (°C)	-5~35			
High Temperature Cut-off (°C)	65			
Reconnect Temperature (°C)	48			



Supports Series Connection

WALL-MOUNTED LiFePO4 BATTERY

CMLB-25.6-200-WM
CMLB-51.2-100-WM



CMLB-25.6-200-WM
CMLB-51.2-100-WM



GRADE A LiFePO4 Cells, > 8000 Cycles @ 50% DoD, 0.5C @ 25°C



Parallel Up to 16 Modules



Supports CAN/RS485, compatible with major inverter brands



WiFi & Bluetooth for real time monitoring



Fire-redundant wiring and laser welding for enhanced safety



Built-in Smart BMS with multi-protection (OVP, ODP, OTP, SCP)

MODEL NAME	CMLB-25.6-200-WM	CMLB-51.2-100-WM
GENERAL SPEC		
Rated Voltage (V)	25.6	51.2
Rated Capacity (Ah)	200	100
Rated Energy (Wh)	5120	5120
OPERATION PARAMETER		
Voltage Range (V)	22.4-28.8	44.8-56.0
Charge Voltage Range (V)	28.0-28.8	55.2-56.0
Cut-off Voltage (V)	22.4	44.8
Recommend Charge Current (A)	100	50
Max Charge Current (A)	200	100
Max Discharge Current (A)	200	100
PHYSICAL PARAMETER		
Dimension : L*W*H (mm)	455*575*230	550*400*178
Weight (kg)	44.5	44.5
Communication Port	RS485 、 CAN	
Display	LED+LCD	
Terminal	OT terminal	
ENVIRONMENT PARAMETER		
Humidity (%)	5-95% RH	
Charge Temperature Range (°C)	0-45	
Discharge Temperature Range (°C)	-20-60	
Storage Temperature Range (°C)	-20-45	
SERVICE LIFE		
Cycle Life	>8000 (0.5C @ 25°C, 50% DOD)	
Design Life (Year)	15	



MOBILE LiFePO4 BATTERY

CMLB-51.2-230
CMLB-51.2-280



CMLB-51.2-230
CMLB-51.2-280



GRADE A LiFePO4 Cells, > 8000 Cycles @ 50% DoD, 0.5C @ 25°C



Parallel Up to 16 Modules



Supports CAN/RS485, Compatible with Major Inverter Brands



WiFi & Bluetooth for Real Time Monitoring



Fire-redundant Wiring and Laser Welding for Enhanced Safety



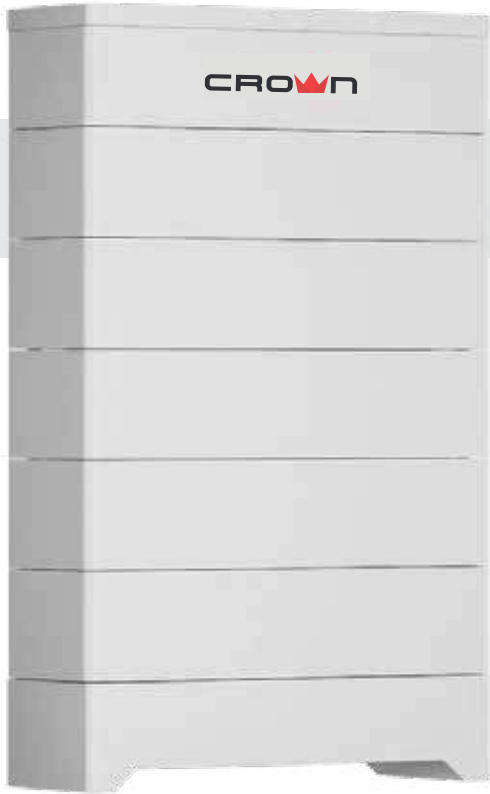
Built-in Smart BMS with Multi-Protection (OVP, ODP, OTP, SCP)

MODEL NAME	CMLB-51.2-230	CMLB-51.2-280
GENERAL SPEC		
Rated Voltage (V)	51.2	
Rated Capacity (Ah)	230	280
Rated Energy (Wh)	11776	14336
OPERATION PARAMETER		
Voltage Range (V)	44.8-56.0	
Charge Voltage Range (V)	55.2-56.0	
Cut-off Voltage (V)	44.8	
Recommend Charge Current (A)	100	
Max Charge Current (A)	200	
Max Discharge Current (A)	200	
PHYSICAL PARAMETER		
Dimension : L*W*H (mm)	455*750*230	460*900*231
Weight (kg)	93.8	124.5
Communication Port	RS485 、 CAN	
Display	LED+LCD	
Terminal	OT terminal	
ENVIRONMENT PARAMETER		
Humidity (%)	5-95% RH	
Charge Temperature Range (°C)	0-45	
Discharge Temperature Range (°C)	-20-60	
Storage Temperature Range (°C)	-20-45	
SERVICE LIFE		
Cycle Life	>8000 (0.5C @ 25°C, 50% DOD)	
Design Life (Year)	15	



HIGH VOLTAGE LiFePO4 BATTERY

CMLB-5.1HV



CMLB-5.1HV

-  20 Stack Parallel
-  9 Series Per Stack
-  Ultra Safe LFP Material
-  High Voltage Battery
-  Short Circuit Protection
-  Flexible Remote Upgrade

MODEL NAME	CMLB-5.1HV
Scalable Connection	2~9pcs series per stack, up to 20 stack parallel
Battery module nominal voltage	70.4V
Battery module nominal capacity	72Ah
Battery module nominal energy	5.069kWh
Protection	IP65
Battery module voltage range	61.6V~80.3V
Recommend charge/discharge current	36A, 0.5C per battery module
Max continuous charge/discharge current	50.4A, 0.7C per battery module
Max continuous charge/discharge power	3.548 kW per battery module
Series per stack	Max. 9 battery series, up to 45kWh per stack
System expansion	10 stack parallel w/o extra device needed, 20 stack parallel with extra BAU installed
DOD	95%
Charging/Discharging temperature	0 ~ +55 °C , -20 ~ +55°C
Recommend storage humidity	10% ~ 90% RH
Warranty *	10 years, 6000 cycles; whichever occurs first
Remote Maintenance	Flexible remote upgrade, Original firmware backup to prevent upgrade failure

*@25°C , 95% DOD, 0.5C testing condition

